

KAVACH AI

India's AI-Powered Digital Safety Shield

Defeating Counterfeiting, Fraud & Digital Arrest Scams

TEAM: BITBUDDIES

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Track: Smart Cities / Public Safety / Digital Trust / Geospatial Law Enforcement

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1. Problem Statement & Executive Context

The Growing Public Safety Crisis:

India registered over **1.14 million cybercrime complaints** in 2023 — a 60% surge from 2022. Among the most dangerous emerging threats are '**Digital Arrest**' scams, where victims are psychologically held hostage over video calls by criminals impersonating CBI, ED, or Police officers. The Ministry of Home Affairs reported citizens defrauded of over **Rs 1,776 crore** in the first nine months of 2024 alone.

Concurrently, counterfeit currency (Fake Indian Currency Notes — FICN) continues to undermine financial systems, with high-quality Rs 500 fake notes bypassing traditional manual detection. Existing law enforcement frameworks are reactive, stepping in only post-complaint rather than neutralizing threats at the point of contact.

2. Proposed Solution — KAVACH AI

KAVACH AI is an integrated, multi-agent AI public safety platform designed to equip citizens, financial institutions, and law enforcement agencies (LEO) with real-time proactive tools to detect, disrupt, and neutralize digital fraud and counterfeit currency.

- **Citizen Fraud Shield:** Multilingual conversational advisory in 12 regional languages providing real-time call/message threat verdicts & 1-click reporting to NCRB 1930.
- **Digital Arrest Scam Detector:** ML-powered transcript & speech classifier trained on 15 call categories to detect impersonation, fear tactics, and urgency indicators.
- **Counterfeit Currency Agent:** Computer vision forensics analyzing microprint, intaglio edge sharpness, HSV color profiles, and printer frequency artifacts across denominations.
- **Fraud Network Graph AI:** Graph algorithms (Louvain Community Detection, PageRank) mapping money mule clusters, suspect nodes, and organized campaign infrastructure.
- **Geospatial Crime Hotspots:** Spatial analytics (DBSCAN & Kernel Density Estimation) identifying cybercrime hotspots and guiding police patrol prioritization.
- **LEO Command & Court Evidence:** End-to-end case management generating court-admissible PDF intelligence packages.

3. Hero Feature Showcase — The 3 Specialized Dashboards

KAVACH AI provides customized, role-based interfaces tailored for Citizens, Law Enforcement Officers (LEO), and System Administrators. Each dashboard equips users with specialized workflows and real-time intelligence.

HERO DASHBOARD 1: Citizen Safety Shield & Scan Portal

- **Scam & Call Classifier:** Real-time NLP & ML transcript analysis for Digital Arrest, Fake CBI, OTP Theft, Loan, & Delivery scams.
- **Counterfeit Currency Verification:** 6-layer image checks (resolution, HSV color profile matching, Laplacian edge intaglio sharpness, texture GLCM, geometry, FFT printer artifacts).
- **Media Forensics (Deepfake Vision & Voice):** Noise consistency, frequency domain anomalies, and synthetic text-to-speech voice cloning checks.
- **Multilingual RAG Advisory:** 12 regional Indian languages (Hindi, English, Tamil, Telugu, Bengali, Marathi, Gujarati, etc.) with 1-click NCRB 1930 Helpline reporting.

HERO DASHBOARD 2: Law Enforcement Officer (LEO) Command Center

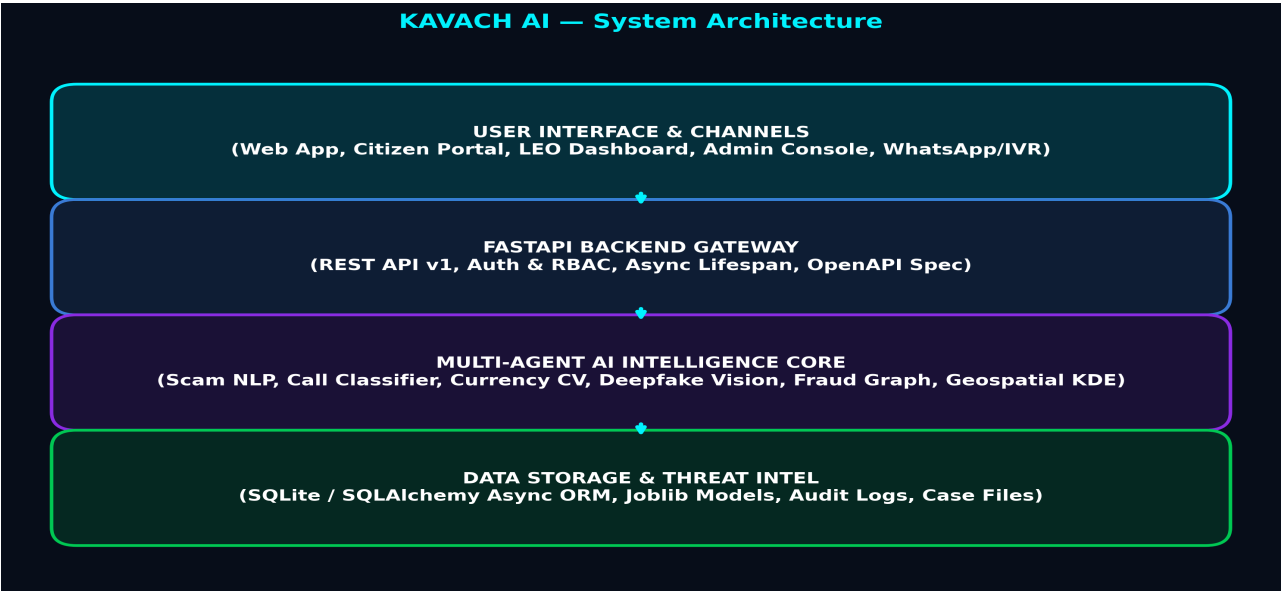
- **Real-Time Case Management:** Case creation, lifecycle status (Open, Investigating, Escalated, Closed), suspect profiling, and evidence attachment vault.
- **Fraud Network Graph AI:** Louvain Community Detection, Degree/Betweenness Centrality, PageRank, and money mule network cluster identification.
- **Geospatial Crime Hotspots:** DBSCAN geographic clustering & Kernel Density Estimation (KDE) density heatmaps for patrol allocation.
- **Court-Admissible Evidence Export:** Automated PDF intelligence package generator for judicial submission.

HERO DASHBOARD 3: Admin Command & Governance Platform

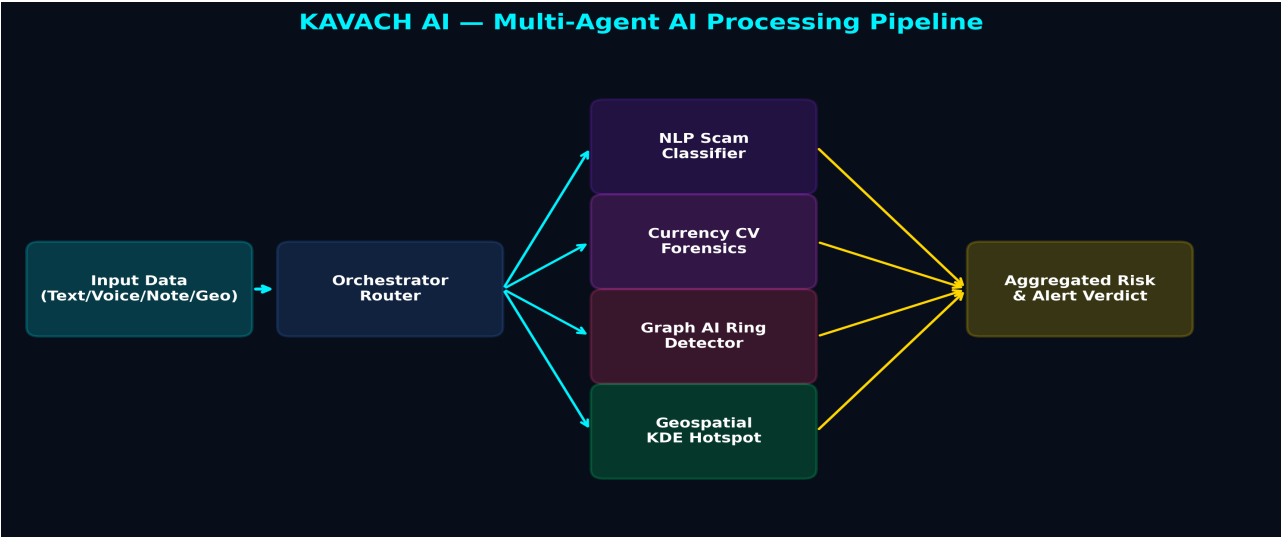
- **User & Role Elevation:** Full user management, role modifications (Citizen → LEO → Admin), credential governance, and badge issuance.
- **Emergency Broadcast Dispatch:** System-wide emergency notification system broadcasting high-priority security alerts to citizens and officers.
- **System Audit Trail:** Real-time audit log inspector capturing every user interaction, API payload, and security event.
- **Health & Performance Diagnostics:** Operational health status of FastAPI backend, database pools, and AI processing modules.

4. System Architecture & Processing Workflows

System Architecture Diagram:



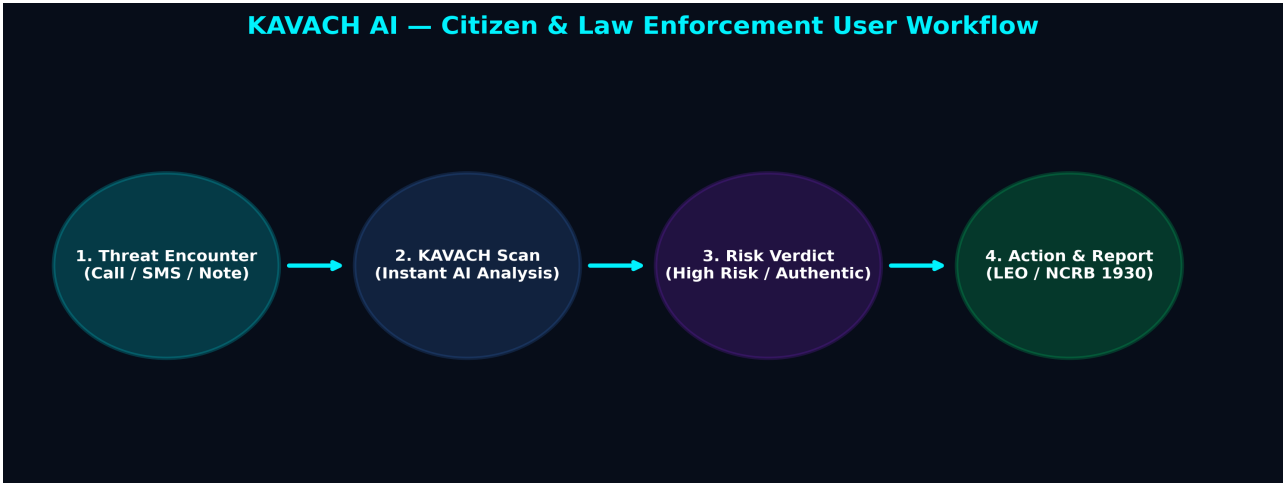
Multi-Agent AI Processing Pipeline:



5. Technology Stack & Technical Matrix

Component	Technologies Used	Key Purpose
Frontend	HTML5, Vanilla JS, TailwindCSS, Iconify, GSAP	Responsive Citizen, LEO & Admin Dashboards
Backend API	FastAPI, Uvicorn, Python 3.11, Pydantic v2	Async REST APIs & Microservices Layer
AI & ML Models	Scikit-Learn, NetworkX, SciPy, NumPy, Joblib, Whisper	Scam NLP, Call ML, Graph AI, Geospatial
Computer Vision	OpenCV, Pillow (PIL), FFT Frequency Analysis	Currency Note Forensics & Deepfake Detection
Database & Auth	SQLite, SQLAlchemy Async ORM, PyJWT, Bcrypt	Persistent Data & Role-Based Access

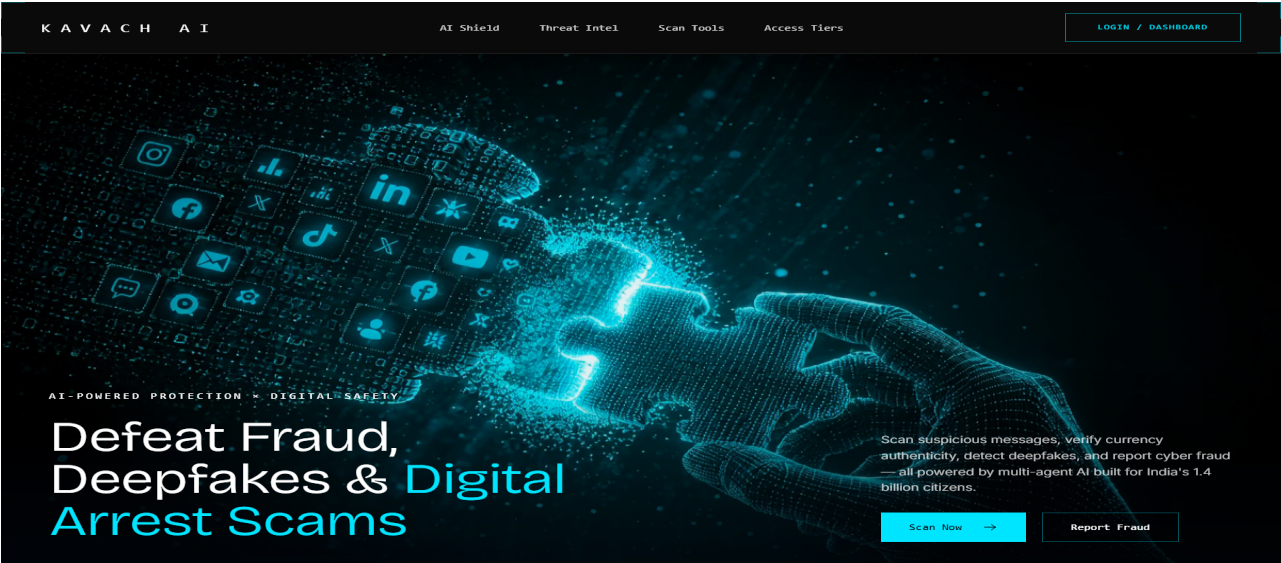
User Processing Journey Workflow:



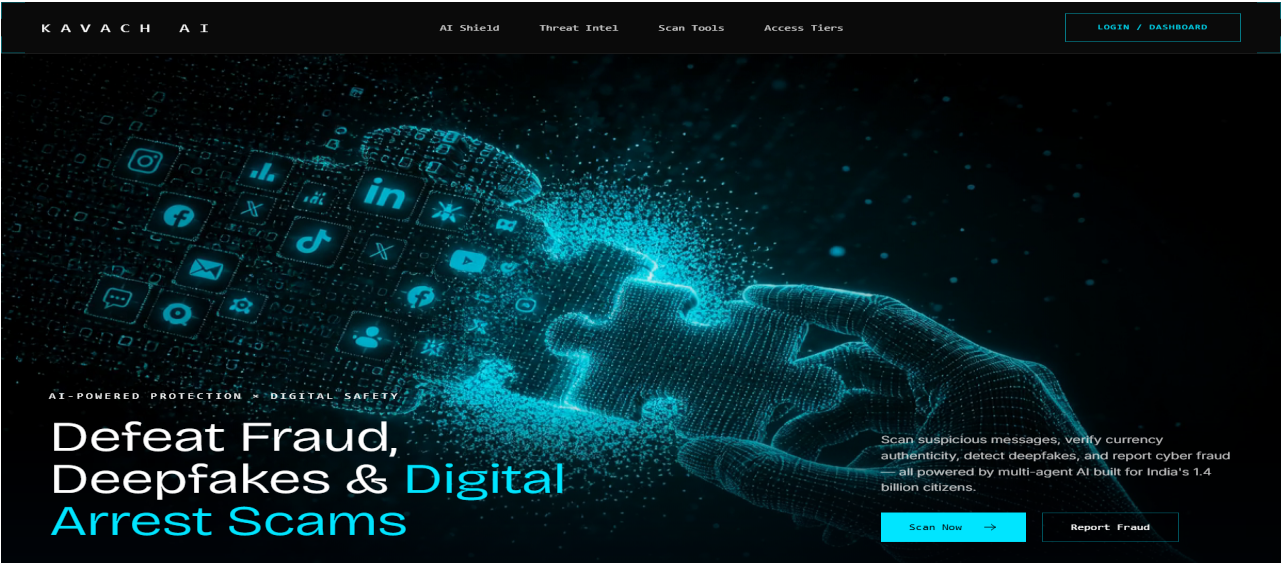
6. Dashboard Overview & Live Screenshots

Actual runtime screenshots captured directly from the live local execution environments (Port 3000 & Port 8000).

KAVACH AI Public Landing Page & Citizen Threat Shield:



Integrated Command Dashboard & Multi-Agent Analytics:



7. Future Scope & Strategic Roadmap

- **WebRTC Live Call Interception:** Integrating real-time audio stream monitoring directly on mobile devices to warn citizens mid-call during active digital arrest attempts.
- **Direct Carrier & WhatsApp Cloud API Integration:** Webhook bridges connecting telecom service providers and Meta WhatsApp Business API for instant threat blocking.
- **Distributed Graph & Database Scaling:** Upgrading SQLite persistence to enterprise PostgreSQL + Neo4j for multi-billion transaction graph analysis.
- **Hardware UV Sensor Integration:** Calibrating computer vision models for specialized UV/IR camera hardware on banking note counting machines.

8. Socio-Economic Impact & Evaluation Alignment

KAVACH AI addresses the core evaluation criteria of the ET AI Hackathon 2026:

- **Innovation (25%):** Multi-agent fusion uniting NLP, Computer Vision, Graph AI, and Geospatial analytics into a single public safety shield.
- **Business & Social Impact (25%):** Safeguarding citizen life savings against psychological digital arrests and preventing counterfeit currency proliferation.
- **Technical Excellence (20%):** Self-contained, zero-external-dependency production architecture with 25+ REST endpoints and trained ML models.
- **Scalability & UX (30%):** Async FastAPI backend, 12 regional languages, and intuitive dark-mode glassmorphism interface.

SUBMISSION SIGN-OFF — TEAM BITBUDDIES

Project **KAVACH AI** is fully operational locally on **http://localhost:3000** (Frontend) and **http://localhost:8000** (Backend API). All code, trained models, and dataset assets are committed and backed up on GitHub: <https://github.com/Jagrati850/KAVACH-AI>.